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Tailored High Expendable Plug for High-Temperature, High-Production Wells to Mitigate Operational Losses in Workover Operation

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Abstract

In oil and gas workover operations, maintaining the integrity of well barriers is important to prevent potential well control incidents. These barriers incorporate mechanical well barrier and fluid barrier (kill fluid), collectively ensuring safe operational conditions. Historical drilling cases face the challenges posed by high permeability reservoirs, which have necessitated prolonged effort to cure the losses and activation of Pressurized Mud-Cap Drilling (PMCD). To mitigate such risks, incorporating retrievable nippleless plugs as mechanical barriers emerges as a viable strategy to safeguard against total losses during operations. This abstract discusses a case of workover high temperature carbonate well with high formation losses and challenge to mechanically isolate the reservoir through restricted ID in the well.

This one well in Sarawak water, operating at 365°F, encountered tubing-casing communication issues, which led to the well-being suspended due to concerns of well integrity. As a short-term solution, an expandable steel patch was installed across the tubing leak, which reduced the internal diameter (ID) clearance. This constraint posed a significant challenge for deploying standard intervention tool or plugs capable of passing through the restricted ID.

To address these challenges, a novel approach was proposed: developing a customized high expansion retrievable nippleless plug with a smaller outer diameter capable of withstanding high temperatures. Additional features included a junk catcher with funnel design connected at the top to prevent debris accumulated around the plug element and a junk extension to capture debris falling into the plug, mitigating any risk on retrieving the plug later. This plug was also equipped with a multi-cycle interventionless glass barrier at the most bottom, serving as a temporary solution for plugging, providing bi-directional barrier capabilities. The glass component was designed to shatter after a pre-defined number of pressure cycle which can eliminate the need to run a prong to equalize the plug before retrieving it and fasten up the equalising process. This innovative solution represented a pioneering effort, necessitating rigorous testing to ensure it complied with international standards (ISO 14310 V0) for barrier integrity, load and pressure, as well as operational reliability in extreme thermal conditions (ISO 14998:2013/V0).

The customized high expansion retrievable plug successfully isolated the production zone, enabling subsequent workover operations to proceed smoothly without encountering losses. This outcome not only

prevent the need for costly PMCD but also demonstrated cost savings and operational efficiencies in similar high-temperature well scenarios.

In conclusion, this case study shows the importance of adaptive engineering and innovation in addressing complex challenges in oil and gas operations. By utilizing advanced materials and design principles, operators can minimize operational risks, and optimize costs associated with well control and intervention measures. This abstract encapsulates where technical achievements meet practical implications for safe and efficient workover practices on high-temperature and high producer well.

Introduction

Workover operations play a pivotal role in sustaining production and prolonging the lifecycle of oil and gas wells. In mature fields— particularly in carbonate reservoirs—these operations become increasingly complex due to aging well infrastructure, high formation permeability, natural fractures, and elevated bottomhole temperatures. Ensuring well integrity during such activities is essential to prevent uncontrolled flow, ensure personnel safety, and maintain reservoir access.

In conventional practice, well integrity during workover is maintained using a combination of fluid-based barriers (such as kill-weight brines) and mechanical solutions like retrievable packers or bridge plugs. However, in highly permeable or naturally fractured carbonate formations, these fluid barriers are often ineffective due to uncontrollable losses into the formation. To address this, operators may resort to pumping Lost Circulation Materials (LCM)—such as fiber-based or particulate blends—aimed at bridging fractures or sealing loss zones.

While LCMs may offer temporary control, they come with trade-offs. In carbonate formations, where productivity often depends on vugs and fracture networks, LCM invasion can plug the very pathways that enable production, leading to irreversible near-wellbore damage. Moreover, once deployed, these materials are not retrievable and can complicate or prevent future interventions. The use of cement squeezes or chemical seals, although effective in some wells, can similarly compromise reservoir deliverability and limit long-term production potential.

These risks are even more critical in high gas production wells, such as the one in this case located in the Sarawak region. With production potential exceeding 200 million standard cubic feet per day (MMscf/d), protecting the reservoir from damage is of utmost importance. Any formation impairment could result in substantial revenue loss, lower recovery efficiency, and limited intervention flexibility in the future.

These limitations underscore the need for retrievable mechanical isolation solutions—especially in wells where permanent sealing is undesirable. A mechanical plug can provide temporary, gas-tight isolation without introducing solids or chemicals into the formation. More importantly, it can be easily retrieved after the intervention is complete, preserving the integrity of the wellbore and the productivity of the reservoir.

This paper presents a real-world application of such a mechanical solution in a high-temperature offshore carbonate well in Sarawak. Following the identification of tubing-casing communication and the installation of an expandable steel patch—which significantly reduced internal diameter (ID)—standard intervention tools were no longer viable. In response, a customized high-expansion retrievable plug was engineered, qualified, and deployed to enable continued workover operations without risking further formation damage or production impairment.

Background and Problem Statement

The subject well is a high-rate carbonate producer located in a field offshore Sarawak. Historically, the reservoir exhibited high permeability and extensive natural fracturing, contributing to severe lost circulation during drilling and well intervention phases. Following a routine annulus pressure test, the well showed signs of tubing-casing communication, triggering integrity concerns. Further diagnostics confirmed a tubing leak near the production zone.

To quickly re-establish a pressure barrier, an expandable steel patch was installed across the leaking section. While the patch temporarily restored well integrity and halted annular pressure buildup, it significantly reduced the internal diameter (ID) of the tubing. The remaining ID, measured at approximately 5.497 inches, restricted access for standard intervention tools including conventional retrievable plugs, perforation assemblies, and pressure gauges.

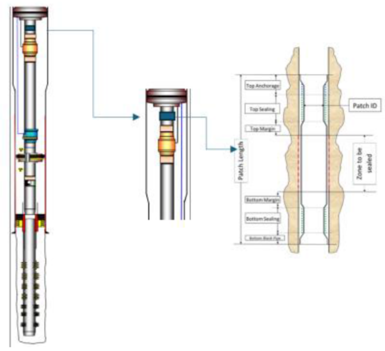

Figure 1—The location of the expandable steel patch

Complicating the issue further, the well operates in a high-temperature environment, with a bottomhole static temperature of 365°F (185°C). This exceeds the maximum continuous service temperature for many standard elastomers and limits the selection of available mechanical plug systems.

Historically, this reservoir had posed severe loss control challenges, with prior drilling operations requiring Pressurized Mud-Cap Drilling (PMCD) to maintain hydrostatic balance. Any failure to fully isolate the zone during workover could result in similar loss scenarios, with significant implications for safety, environmental risk, and cost.

Faced with these constraints, the operator sought a mechanical barrier that could:

1. Pass through a severely restricted ID,
2. Withstand HPHT conditions,
3. Provide bi-directional sealing,
4. Be retrieved without mechanical equalization, and
5. Be deployed without a conventional rig.

The complexity of these combined requirements eliminated the feasibility of commercial plug systems, necessitating a fit-for-purpose solution through custom tool design and accelerated engineering validation

Engineering Objectives

In light of the unique challenges posed by this well—restricted internal diameter, high bottomhole temperature, well integrity concerns, and the need for wireline deploy-ability—a comprehensive set of engineering objectives was established to guide the design, development, and validation of the custom plug solution.

a. **Dimensional Compatibility**
   The retrievable plug had to traverse a 5.497″ restricted ID while still being able to expand and seal within a 7.0-inch casing. This required an exceptionally compact run-in profile combined with a high-ratio expansion mechanism that could anchor securely and deliver reliable sealing performance.
b. **High-Temperature Endurance**
   All materials, including elastomers and structural components, needed to be rated for continuous exposure at 338°F (170°C) which is the temperature at target set depth. This ruled out standard nitrile

or hydrogenated elastomers and required selection of HPHT-qualified polymers and high-strength corrosion-resistant alloys.

c. **Bi-Directional Barrier Integrity**

The tool had to provide gas-tight sealing in both directions (upward and downward pressure differential), with a validated differential pressure rating of 7,500 psi, meeting the criteria of ISO 14310 V0 for mechanical barriers. This was essential to ensure safety and well control during fluid displacement, pressure testing, and [intervention].

The Plug assembly was tested according to ISO 14310:2008/V0 with a maximum differential pressure of 7500 psi (517 bar) from top and bottom and a temperature range of 170.6-100°C in a 159.4/156.6 mm (7″ 29 ppf) casing. The results showed the following:

- Recorded leakage was 0 bubbles for all cycles.
- The Plug was pulled without difficulties through a 5.475″ restriction.
- All components were found to be conforming to serviceable standards.
- The results of this test meet the requirements for the standard design validation grade V0 described in the standard ISO 14310:2008.

d. **Passive Equalization System**

Standard retrievable plugs often require a prong or mechanical equalization tool to relieve trapped pressure before pulling. However, given the access constraints and potential for misalignment, the new design had to incorporate a passive equalization feature—specifically, a multi-cycle interventionless glass barrier engineered to shatter after a pre-defined number of pressure cycles (e.g., 5-7), eliminating the need for separate mechanical runs.

e. **Debris Management**

The risk of scale or steel debris—especially from the previously tubing cut above plug—posed a serious threat to plug setting, sealing, and retrieval. The plug had to include a funnel-shaped inlet to divert debris, and a junk extension below the packer to trap solids and prevent accumulation around sealing elements.

f. **Wireline Deployment**

The entire system had to be deployable via wireline without requiring a full workover rig. This was critical to reduce costs and minimize HSE exposure.

g. **Accelerated Development Timeline**

Given operational urgency, the tool had to be designed, manufactured, tested, and deployed within 10 weeks. This necessitated a modular approach using existing qualified components (where possible), rapid prototyping, and real-time collaboration between operator, vendor, and field teams.

These objectives formed the foundation of the tool development process and ensured the final design would meet both performance standards and field-operational practicality

Solution Design and Development

To address the unique operational constraints, a custom high-expansion retrievable nippleless plug was conceptualized and engineered. The tool design was based on modular principles, incorporating proven components adapted for a restricted ID and high-temperature environment. Key elements of the development process included advanced mechanical design, material selection, performance modelling, and rigorous qualification testing.

a. **Mechanical Design Architecture**

The core architecture of the plug was designed around a wedge-driven expansion mechanism. This system enabled the plug's packer element and anchoring slips to be compressed during deployment and then expanded radially with controlled force at depth. The design allowed a maximum run-in OD of 5.35 inches with expansion ratio up to 6.293 inches. Expansion modelling using finite element analysis (FEA) was performed to confirm contact pressure, sealing performance, and radial stress distribution under differential pressure loads.

Key components of the design included:

- **Top Junk Catcher with Funnel**: Prevented debris from accumulating around the plug element for easier retrieval later.

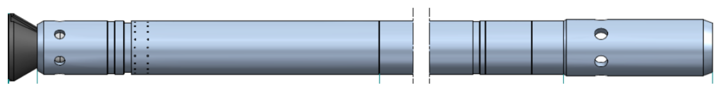

Figure 2—Junk catcher with Funnel

- **Elastomeric packer**: Rated to 338°F, it was designed for long-term exposure under thermal cycling without loss of shape memory or integrity and tested to Max API ID of 7″, 29ppf.
- **Slips assembly**: Components machined from Super 13Cr NACE and Inconel 718 to maintain anchoring under thermal expansion, contraction and sour environment. Slips module is supported by anti-extrusion design to withstand bi-directional pressure.
- **Patented Packer Backup System**: Innovative seal module which both compresses and constrains the element, enabling it to withstand extreme differential pressures.

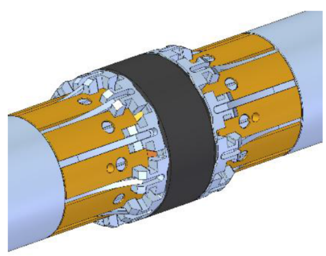

Figure 3—Patented Packer Backup System

- **Junk extension**: Located beneath the packer to trap solids falling from above and to keep the fish neck clear for eventual retrieval.

b. **Passive Equalization System**

A key innovation in the tool design was the use of a multi-cycle interventionless glass barrier. This component served as a temporary pressure barrier once the tool was set. The glass barrier was designed to hold high pressure for a predefined number of cycles (e.g. 5-7). Once completed the pressure cycles, the glass will shatter into very fine particles, there by equalizing the plug. This eliminated the need for a prong or mechanical equalization run—an essential feature for tight-access wells where alignment is uncertain and space is limited.

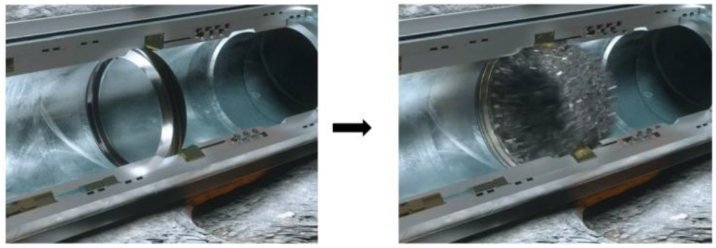

Figure 4—Glass barrier shattered to very fine particles once completed pressure cycle

c. **Material Selection**

Due to the harsh downhole conditions, all materials were selected based on their resistance to:

- Thermal degradation
- Sour gas exposure (H_2S-resistant)
- Scale and debris erosion
- High axial and radial loads

The tool body, slips, and stress-bearing components were constructed from Super 13Cr NACE, Inconel 718 and SS2511, while elastomers were selected from Engineering thermoplastic and other HPHT-compatible polymers.

d. **Testing and Qualification**

Prior to field deployment, the prototype plug underwent full qualification testing simulating downhole environments and field conditions. All tests were completed in accordance with ISO 14310 V0 and ISO 14998:2013 standards, with QA/QC oversight from both the operator and the vendor.

The successful qualification of the plug demonstrated its readiness for deployment in the high-risk Sarawak well. Three units were manufactured, with one deployed, one retained as backup, and one kept for further benchmarking and potential field trials

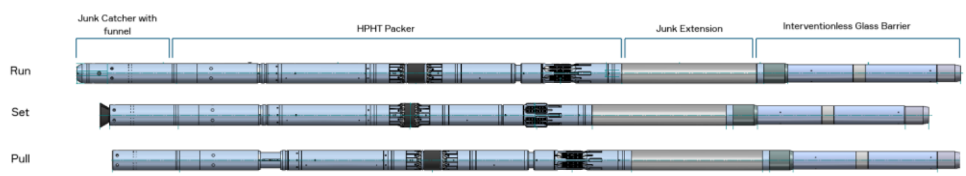

Figure 5—Nippleless plug combination of Junk catcher with funnel, HPHT packer, Junk Extension and Interventionless Glass Barrier with max running OD of 5.35 inches

Operational Deployment

Following successful qualification, deployment was executed using a wireline unit mounted on a rig. This allowed for precise tool control while minimizing operational cost and footprint, especially important in offshore brownfield settings.

a. **Pre-Deployment Preparation**

Prior to running the plug, the following preparatory steps were taken:

- Gauge Ring Run: A dummy tool with the same outer diameter (OD) as the custom plug was run to confirm unobstructed passage through the previously installed steel patch. This verified that the restricted ID section would not impede deployment.

- Tool Readiness and Testing: The Junk Catcher, HPHT Plug, Junk Extension assembly and Interventionless Glass Plug was successfully tested in Norway as per ISO 14310 V0. The setting tool was chosen to apply the sufficient force required to actuate the axial force to set the plug without exceeding mechanical tolerance of the surrounding tubulars.
- Contingency Plans: Retrieval procedures and backup tools were staged on standby, and a risk mitigation matrix was developed to address possible setting failure, equalization delays, or debris-related obstruction.

b. **Deployment Sequence**

The plug was run in-hole using wireline, with depth correlation achieved via casing collar locator (CCL) and gamma-ray logging. The setting depth was selected based on reservoir isolation requirements and clearance from the patch and completion accessories.

Key execution steps included:

1. Tool Descent: Controlled run-in to minimize impact against patch ID edges.
2. Set Confirmation: Uses Hydrostatic Setting Unit (HSU) to set the plug. The HSU Trigger Mechanism is activated by electric current on E-line and utilizes in situ well pressure for generating piston force to produce stroking force. A clear surface indicator confirmed the tool was anchored. The HSU logging feature enables validation of successful setting profile of the set plug.
3. Post-Set Pressure Test: The well was pressured up to 2,500 psi to verify initial sealing integrity, followed by a hold test at 3,000 psi for 15 minutes. No pressure drop was observed.

c. **In-Well Performance**

The well was placed in suspended mode for 21 days, during which time multiple operations were conducted, including:

- Fluid displacement and circulation,
- Tubing leak remediation activities,
- Diagnostic pressure tests.

The plug underwent a total of seven pressure cycles, each with varying hold durations and pressure build-up rates. The glass barrier shattered as designed after the seventh cycle, allowing pressure equalization across the plug.

d. **Retrieval Operations**

Following glass plug equalization:

- The plug was pulled via wireline in a single retrieval run.
- No mechanical resistance, overpull, or fish were encountered.
- Post-job inspection confirmed no scale bridging or junk interference.
- The elastomeric element and slips were intact and showed minimal deformation, validating their design under thermal and pressure cycling conditions.

This deployment marked the first field application of the tool in Malaysia, and its flawless performance under restricted ID and HPHT conditions reinforced its potential as a robust barrier solution for similar wells across the operator's portfolio.

Results and Impact

The successful deployment and retrieval of the customized high-expansion plug delivered significant technical, operational, and financial outcomes. It proved not only the viability of the tool's design under extreme conditions but also demonstrated the broader value of adaptive engineering in field operations.

a. **Technical Outcomes**

- Full Zonal Isolation Achieved: The plug effectively isolated the production formation throughout the 21-day suspension window, with no pressure communication, leak paths, or fluid losses observed.
- ISO V0 Compliance Maintained: Pressure tests confirmed the tool met ISO 14310 V0 barrier integrity standards, sustaining up to 7,500 psi differential pressure without gas or fluid leakage in either direction.
- Thermal and Pressure Durability Validated: The tool performed continuously under 338°F (170 Deg C) temperature at setting depth and endured seven full pressure cycles without degradation of materials, seals, or anchoring components.
- Retrievability Confirmed: The passive glass plug equalization system worked flawlessly, allowing for safe plug recovery via wireline without the need for mechanical intervention tools or pressure snubbing.

b. **Operational Benefits**

- Avoidance of PMCD: The plug enabled a mechanical isolation solution, eliminating the need for Pressurized Mud-Cap Drilling (PMCD), which would have introduced considerable cost and complexity.
- Minimized Equipment Footprint: By allowing slickline deployment, the solution avoided rig mobilization, reducing offshore footprint, exposure hours, and logistics burden.
- Improved Safety and Simplicity: No hot pipe, heavy lifting, or rotary drilling systems were required. The plug's design eliminated prong usage, which often introduces complexity and mechanical risk, especially in restricted wells.
- Time Savings: The deployment and retrieval were completed in less than 3.5 hours each, compared to typical multi-day timelines for conventional plug-setting and equalization methods under constrained conditions.

c. **Cost Avoidance and Value Creation**

- Direct Cost Avoidance: The estimated cost of executing PMCD for this well—including logistics, equipment, fluids, and lost time—was between USD 750,000 and 1 million. These costs were entirely avoided through the use of the custom plug.
- Deferred Production Recovery & Revenue Upside:
  Following successful isolation and remediation of the tubing leak, the well was returned to production. Flowrate increased significantly from 110 million standard cubic feet per day (MMscf/d) to 200 MMscf/d.
- Strategic Value: Beyond the direct financial return, the success of this intervention also:
  - Prevented production deferment,
  - Avoided the need for recompletion or sidetrack,
  - Preserved wellbore accessibility for future interventions,
  - Qualified the technology for similar constrained HPHT wells.

Best Practices and Lesson Learnt

The deployment of the customized high-expansion retrievable plug offered a number of technical insights and practical lessons that are directly applicable to similar well scenarios, particularly in mature, thermally stressed assets.

a. **The Importance of Custom Engineering**

One of the key takeaways from this case was the importance of tool design customization based on actual well constraints rather than forcing a standard product into a non-standard scenario. The plug was purpose-built to handle the precise combination of:

- Restricted ID,
- High temperature,
- High pressure,
- Debris presence, and
- Retrieval without mechanical equalization.

This tailored approach, while initially requiring more effort and coordination, eliminated risks and inefficiencies that would have been introduced had a commercial off-the-shelf solution been used and modified in the field.

b. **Proactive Vendor-Operator Collaboration**

Rapid design, testing, and deployment were possible only because of close collaboration between the operator, tool manufacturer, field engineers, and QA/QC representatives. Weekly reviews, live data sharing, and co-managed testing ensured alignment from design intent to downhole performance.

This project proved the value of cross-functional, agile engineering over the traditional linear design-manufacture-deploy cycle.

Accelerated field-readiness was achieved in less than 10 weeks, compared to typical development timelines of 6-9 months for similar tools.

c. **Qualification Testing Must Simulate Field Conditions**

Laboratory qualification that simulates realistic downhole conditions—such as thermal cycling, pressure reversals, and debris exposure—was key to successful deployment. Too often, tools are tested under idealized conditions that fail to account for scale, corrosion, and thermal expansion.

The debris flow simulation and post-set thermal stress testing conducted for this tool directly translated to downhole reliability. The packer and anchoring system performed as predicted, even after 21 days of exposure to dynamic pressure and temperature cycles.

d. **Simplified Equalization Enhances Retrieval Confidence**

The use of a multi-cycle glass plug was a simple yet powerful innovation. Traditional prong tools for equalization can introduce additional rig-up complexity, misalignment risk, and mechanical failure potential—especially problematic in HPHT environments with limited access.

By replacing this with a passive, pressure-responsive system, the design enhanced retrieval safety and simplicity. Operators expressed greater confidence in recovery operations, knowing that no separate equalization run was needed.

e. **Modular Tool Design Enables Broader Applications**

The modular nature of the plug—built around a core high-expansion mechanism but adaptable with different subcomponents— means that it can be easily customized for:

- Different ID constraints,

- Different temperature ratings,
- Optional features (e.g., dissolvable elements or telemetry sensors).

This flexibility creates an inventory of adaptable solutions for brownfields, avoiding the "one-tool-fits-all" limitation common in plug-and-packers inventories.

These lessons provide valuable inputs for future planning, especially for interventions in aging assets, carbonate formations, or HPHT wells. The case demonstrates how applied engineering and field-focused innovation can turn seemingly high-risk wells into recoverable and manageable opportunities

Conclusion

This case study has demonstrated that custom-engineered mechanical barrier systems can provide a highly effective solution for complex intervention challenges in high-temperature, restricted-ID carbonate wells. The plug system described here—engineered to pass through a 5.497-inch ID and expand to isolate a 7.0-inch zone at 338°F—enabled the safe and efficient execution of remedial operations without the need for Pressurized Mud-Cap Drilling (PMCD) or a full workover rig.

The key success factors included:

- **Purpose-driven design**, incorporating high-expansion geometry, high-temperature materials, and a debris-managed sealing system.
- **Innovation in equalization**, replacing traditional prongs with a pressure-cycle actuated glass plug to reduce complexity and enhance safety.
- **Full qualification testing**, simulating field conditions for thermal, pressure, load, and debris challenges, in accordance with ISO 14310 V0 and ISO 14998:2013 standards.
- **Operational simplicity**, with successful wireline deployment and retrieval, executed in hours rather than days.
- **Significant cost savings**, with direct avoidance of up to 15% cost from alternative intervention and deferred production costs.

Most importantly, this case illustrates how cross-functional collaboration between operator and service provider—combined with fast-tracked engineering, field-informed testing, and operational discipline—can deliver practical, high-impact results. It also underscores the importance of moving beyond one-size-fits-all tools, especially in mature assets with legacy constraints.

The custom high-expansion retrievable plug has since been qualified for use across other HPHT wells within the operator's portfolio and serves as a template for barrier design in similarly constrained operations.

As the industry continues to extend the life of aging wells while managing higher integrity risks, solutions like this will become increasingly relevant—not only in Malaysia, but globally—where intervention feasibility often hinges on the ability to innovate around mechanical limitations.

Acknowledgment

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